



K.S.RANGASAMY
COLLEGE OF TECHNOLOGY

AUTONOMOUS | TIRUCHENGODE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



AeroLinkers

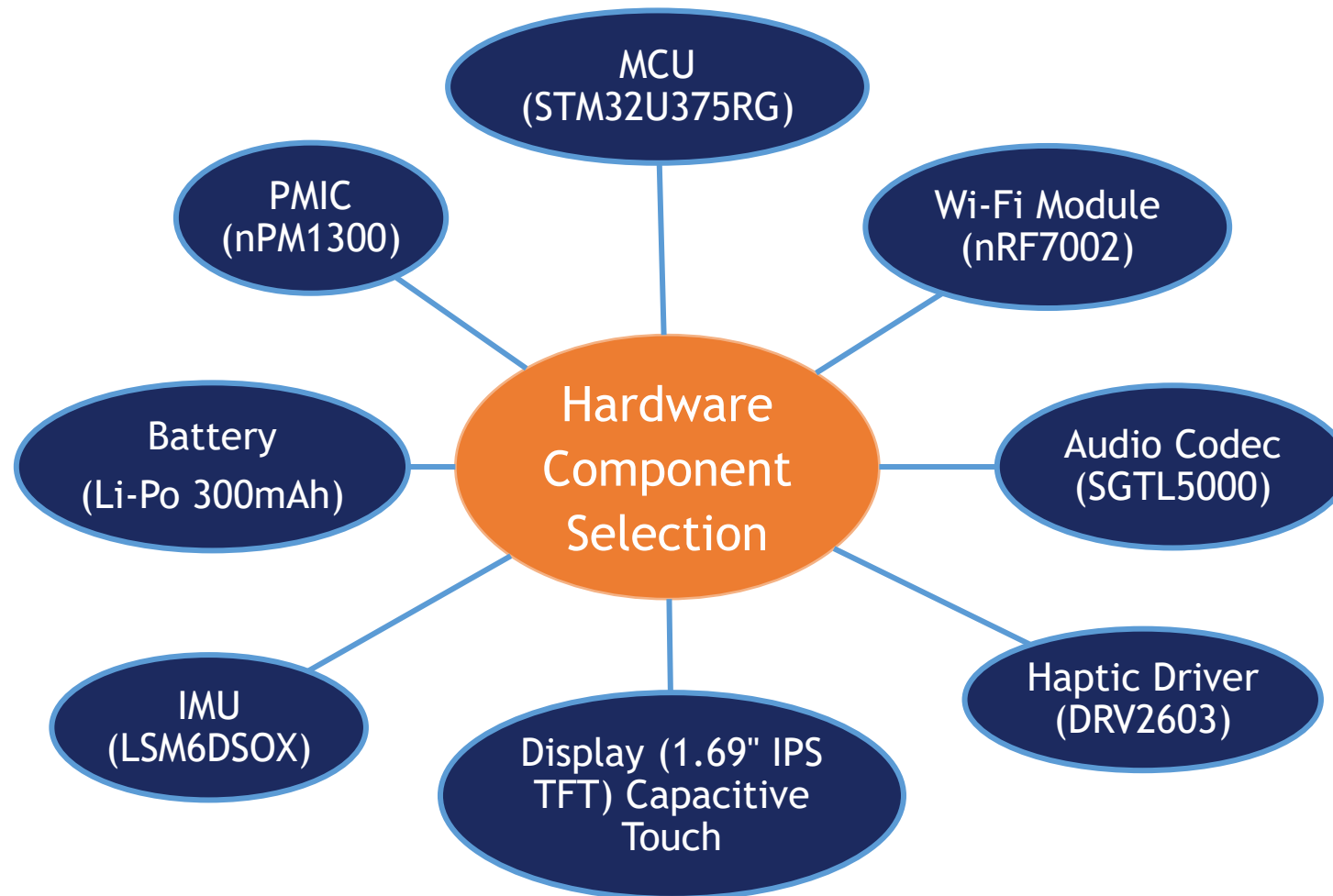


Mr. S. Nithilan, IV-ECE

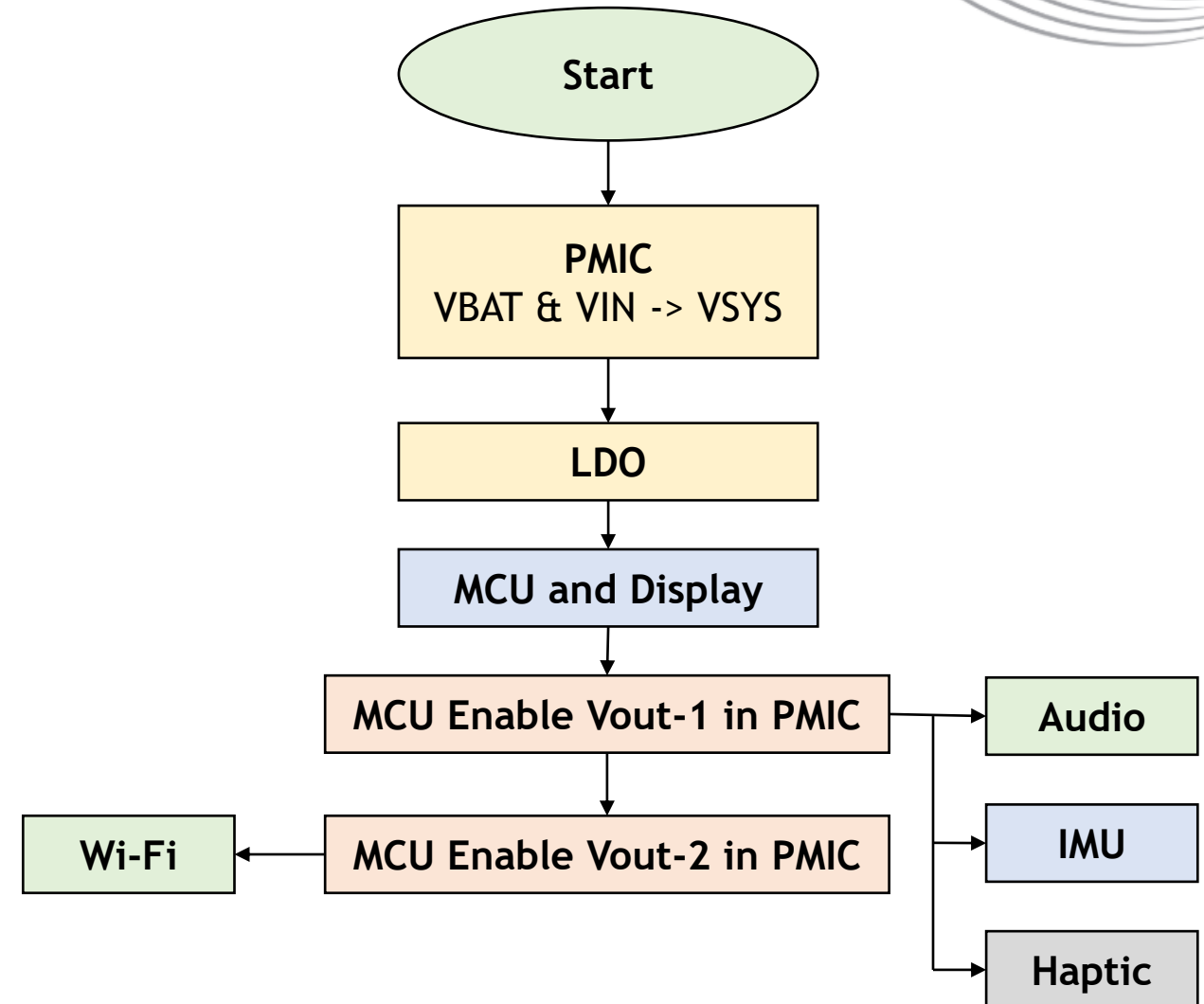
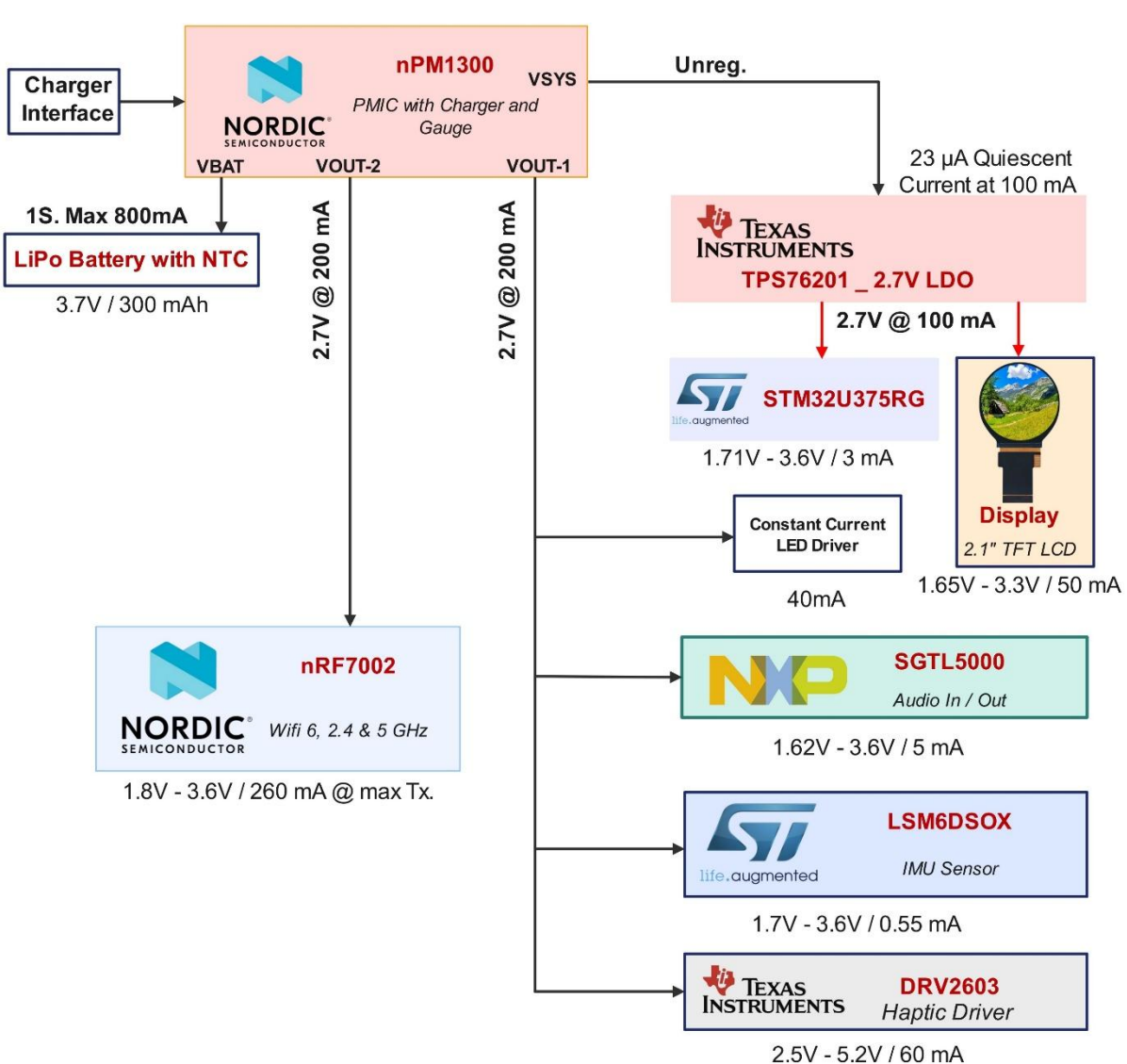
Ms. S. Monisa, II-ECE

Mr. T. P. Nishanth, II-ECE

Components Selection



Power Architecture & Boot Sequence



Power Consumption & Dissipation

All values are in Milliampere (mA)

Operation (Approx. time)	PMIC & LDO	MCU	Wi-Fi	DISPLAY & Back Light (50% PWM)	IMU	AUDIO	HAPTIC (50% PWM)	TOTAL
Booting (5 Sec)	0.823	3	180	50	0.55	5	30	269.37
Data Communication (1 Sec)	0.823	3	60	0	0.55	5	30	99.37
Rest Position	0.823	3	60	0	0.55	0	0	94.37
Wake Position	0.823	3	60	50	0.55	0	0	114.37
Docking	0.823	3	60	0	0.55	0	0	64.37
Vibration (1 Sec)	0.823	3	60	50	0.55	0	30	144.37

Reset Scheme and Interrupt Mapping

- Applications can be reset by User Action Button
- Peripherals can be reset by MCU through its interface
- User Button -> High Priority _ Level-1
- Touch -> Level-2
- Wi-Fi -> Level-3

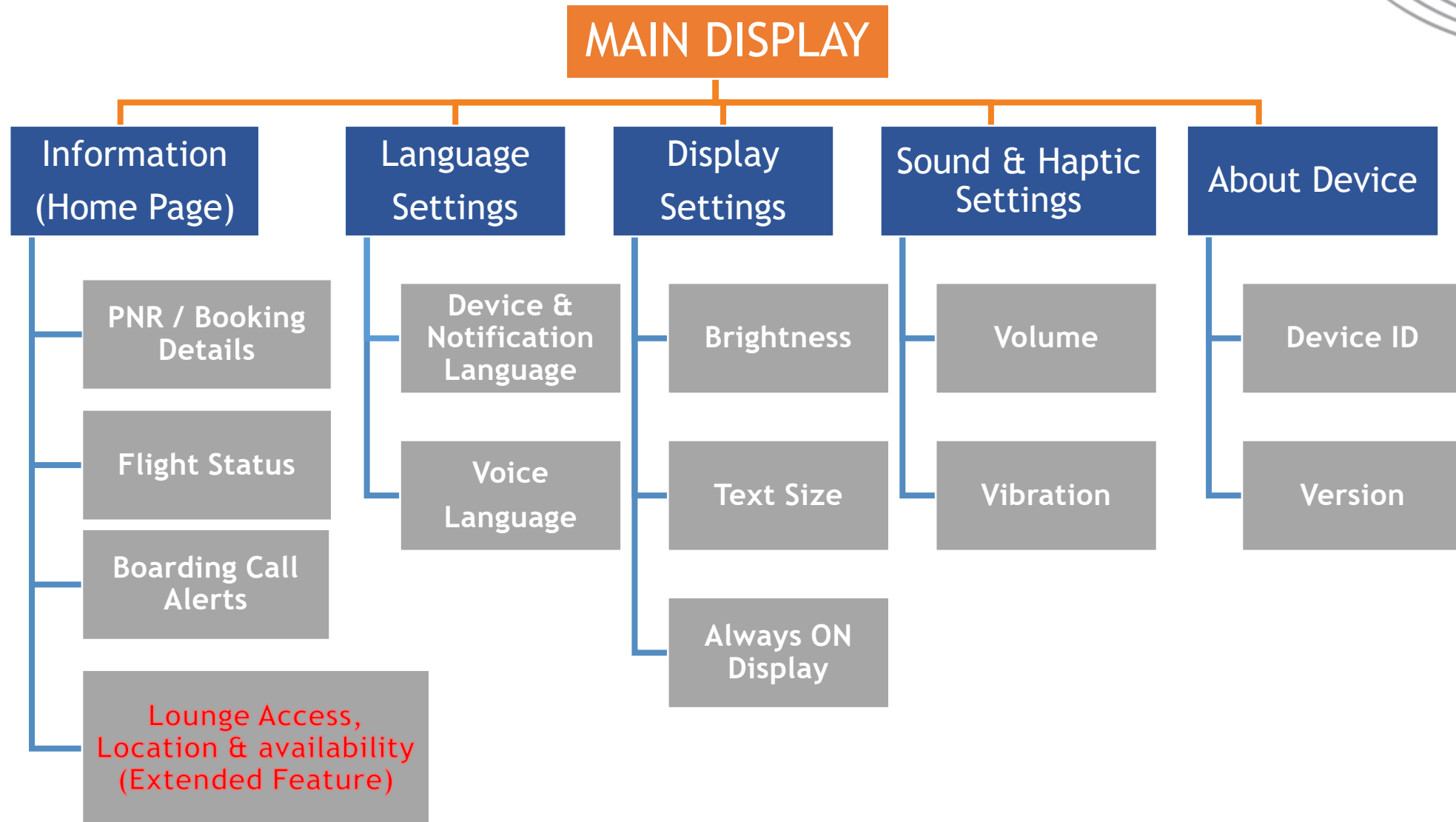
CPU Selection and Memory Requirement

Parameter	STM32U375RG	nRF5340	CC3551E	STM32U5
MCU Type	ULP MCU	Radio-centric SoC	Wi-Fi MCU SoC	ULP MCU
Integrated Radio	None	BLE	Wi-Fi	None
Architecture Impact	Arm® Cortex®-M33 32-bit RISC core	Wireless SoC with two Arm® Cortex®-M33	Arm® Cortex®-M33	Arm® Cortex®-M33
Active Current	16 μ A/MHz	55 μ A	48 μ A	16 μ A
Deep Sleep Current	~1.6 μ A	2.2 μ A	2.8 μ A	1.9 μ A
Lead Time	6-8 wks	10-12 wks	12-14 wks	8-10 wks
Memory (ROM / RAM)	1 MB (1024 KB) / 512 KB	1 MB (1024 KB) / 512 KB	1 MB (1024 KB) / 512 KB	1 MB (1024 KB) / 512 KB

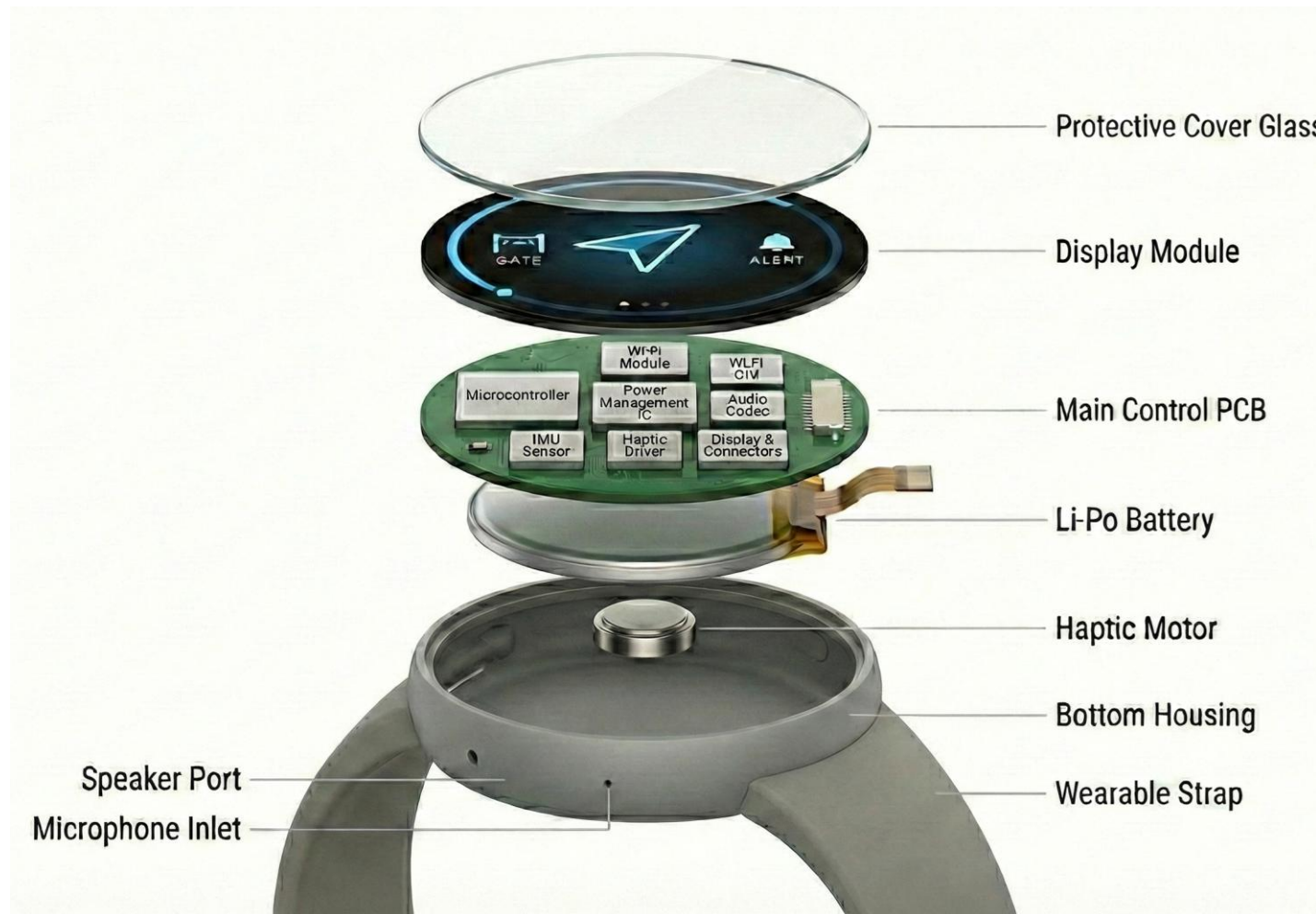
IDE, OS, CLI & Device Driver

- Zephyr RTOS preferred - Low Footprint and Power Consumption
- Linux not preferred - No need of much application and user interface
- Device Driver for all selected components are available in Zephyr support page
- For Command Line Interface, SSH preferred
- WolfSSH library available for establishing SSH in Zephyr

Device Menu



Exploded View





Thank You !



K.S.RANGASAMY COLLEGE OF TECHNOLOGY

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Tiruchengode - 637 215. Namakkal Dt. Tamil Nadu. India

www.ksrct.ac.in

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